nature methods

BRIEF COMMUNICATION

https://doi.org/10.1038/s41592-019-0344-8

SIRIUS 4: a rapid tool for turning tandem mass spectra into metabolite structure information

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Mass spectrometry is a predominant experimental technique in metabolomics and related fields, but metabolite structural elucidation remains highly challenging. We report SIRIUS 4 (https://bio.informatik.uni-jena.de/sirius/), which provides a fast computational approach for molecular structure identification. SIRIUS 4 integrates CSI:FingerID for searching in molecular structure databases. Using SIRIUS 4, we achieved identification rates of more than 70% on challenging metabolomics datasets.

The identification of molecules remains a central question in analytical chemistry, in particular for natural products research, untargeted metabolomics, environmental research, and biomarker discovery. Because of its high sensitivity, mass spectrometry is well suited for the high-throughput characterization of biomolecules. Automated interpretation of tandem mass spectrometry (MS/MS) data is often limited to searching in spectral libraries, so that the user can identify only compounds for which a reference sample has been measured and its spectrum stored. Manual interpretation is cumbersome and work-intense, as current mass spectrometry technology can produce hundreds of thousands of MS/MS spectra per day on a single instrument. Several existing tools for untargeted metabolomics analysis are generalist tools for the complete metabolomics pipeline[1–5].

Here we describe SIRIUS 4, which is a specialized tool that addresses two fundamental questions: What is the molecular formula of the query compound among all molecular formulas, both previously observed and unobserved? And, given a database of molecular structures, what is the structure that best explains the experimental data? Molecular formula identification using isotope pattern analysis was first implemented in SIRIUS version 1, released in 2009 (ref. [6]). In 2011, SIRIUS[2] added methods for the analysis of MS/MS data using fragmentation trees[7]. Version 3.0 of SIRIUS was released in May of 2015; it was a complete rewrite of the software, and implemented the maximum a posteriori estimation from ref. [8], which improved correct molecular formula identification rates from MS/MS data by 2.5-fold. Different from previous versions, version 3.0 did not have a graphical user interface. CSI:FingerID[9] was introduced in October of 2015 and is based on the prediction of a molecular fingerprint of a query compound from its fragmentation tree and spectrum[10]. In contrast to its ancestor FingerID[11], it integrates fragmentation trees in the prediction pipeline, thereby substantially increasing the number of correct identifications. In this first incarnation of CSI:FingerID, users had to submit queries through a web interface. A discussion of related methods is presented in Supplementary Note 1.

On the conceptual level, SIRIUS 4 integrates high-resolution isotope pattern analysis and fragmentation trees with structural elucidation to provide a combined and coherent assessment of molecular structures from MS/MS data for large datasets. Users can now analyze full liquid chromatography–mass spectrometry (LC-MS) datasets, rather than just one spectrum at a time; mass-spectrometry-driven annotations can be obtained for all detected features, not just those passing a preliminary statistical test, say, on fold change. This paves the road to more sophisticated data-mining techniques, exploiting the structure of metabolic pathways[12] or the structural similarity of molecules[13].

On the technical level, SIRIUS 4 and its web service include 213,071 lines of Java code, of which about 94% is new compared with SIRIUS 3.0. CSI:FingerID is seamlessly integrated into SIRIUS 4 via a RESTful (representational state transfer) web service (Fig. 1a), which allows users to process full LC-MS datasets instead of individual compounds. SIRIUS 4 provides an intuitive graphical user interface with six views (Supplementary Fig. 1), including views for CSI:FingerID results and predicted fingerprints. Extensive algorithm engineering and parallelization through a job-scheduling system (Supplementary Fig. 2) have reduced SIRIUS 4 running times by more than two orders of magnitude, and thus reduced the time needed to analyze a full LC-MS run from hours to minutes. SIRIUS 4 integrates a model for scoring isotope patterns that combines absolute and relative noise for peak intensities, and outperforms all previous scorings (Methods). Furthermore, MS/MS spectra with isotope peaks can be processed (Supplementary Fig. 3). Fragmentation tree computation is based on maximum a posteriori estimation from ref. [8] for selection of the molecular formula that best explains the data. SIRIUS 4 also implements automated element detection from isotope patterns[14] using a deep neural network (DNN). The CSI:FingerID web service uses 17 kernels (Supplementary Table 1), of which 10 are new in SIRIUS 4, and predicts more than 1,000 additional molecular properties, mostly from combinatorially generated fingerprints. MS/MS data from 11,728 compounds have been added to the training data (Methods); furthermore, CSI:FingerID can now also analyze negative ion mode data. To score structure candidates, SIRIUS 4 implements the Bayesian network scoring from ref. [15].

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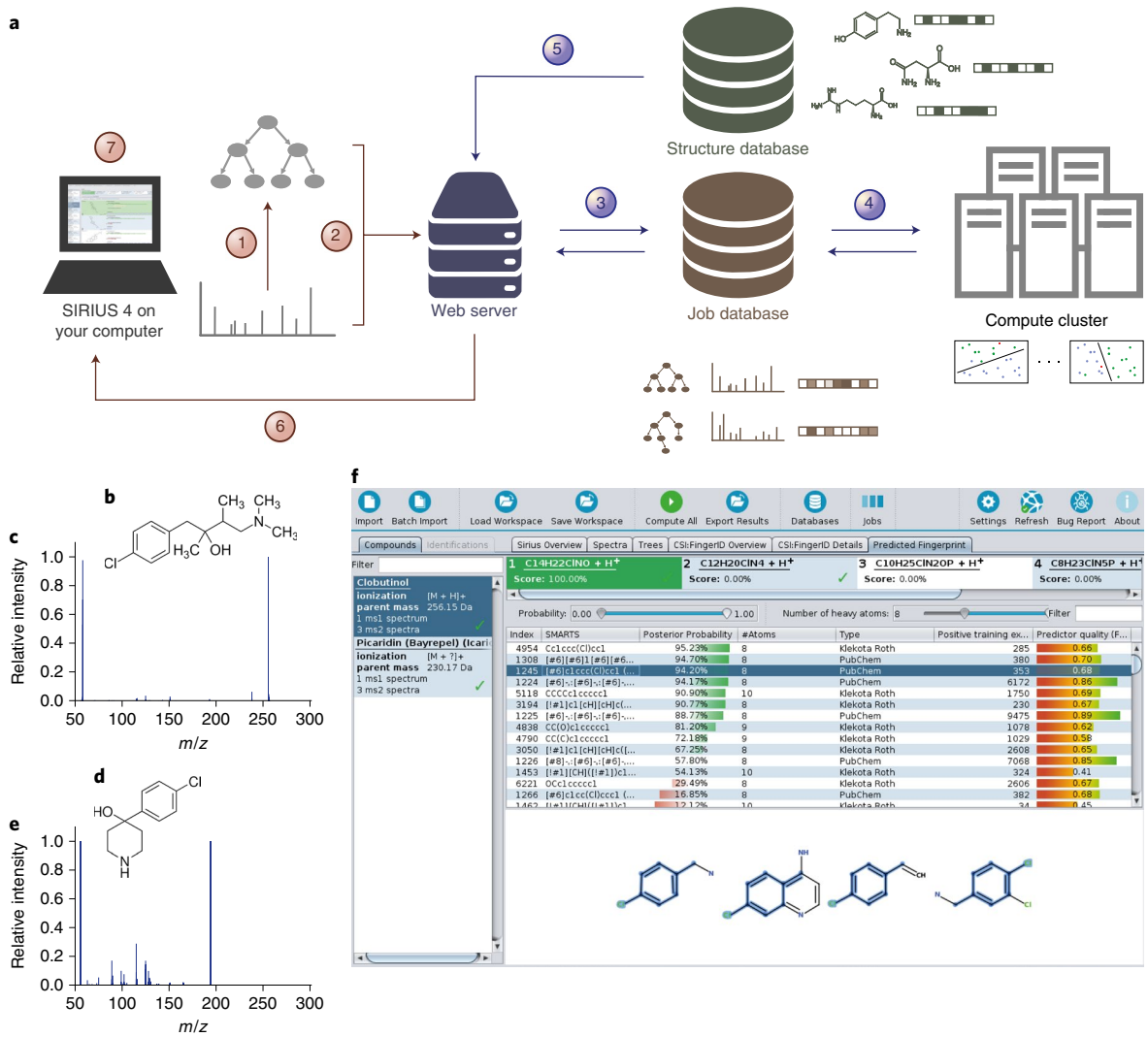

Fig. 1 | Illustration of SIRIUS 4 incorporating CSI:FingerID. a, Integration of CSI:FingerID into SIRIUS 4 via a RESTful web service. Fragmentation trees are computed locally with SIRIUS 4 (1). Fragmentation spectra and trees are uploaded to the CSI:FingerID web service (2). A job is created and temporarily stored in a relational database (3). The compute server fetches each job, predicts the fingerprint, and stores the result in the job database (4). Structures and fingerprints of candidates are retrieved from the structure database (5). The predicted fingerprint plus structures and fingerprints of candidates are returned via the web interface (6). The predicted fingerprint is locally scored against the candidate fingerprints; results are presented in the SIRIUS user interface (7). **b**–**f**, "Predicted Fingerprint" tab. **b**, Structure of clobutinol, not known to SIRIUS 4 and CSI:FingerID. **c**, MS/MS of query clobutinol; this, plus the precursor MS1, is the only input for SIRIUS 4 and CSI:FingerID. **d**,**e**, No MS/MS data for the clobutinol structure are present in the training data of CSI:FingerID (Supplementary Note 2). The structurally closest compound in the training data (measured by PubChem Tanimoto coefficient) is 4-(4-chlorophenyl)-4-piperidinol (**d**) (mass spectrum in **e**). Cosine similarity (**c**,**e**) is less than 0.01. **f**, Predicted fingerprint of the query clobutinol. At this point, no structure database has been searched. Only molecular properties with at least eight heavy atoms are displayed. One molecular property that is predicted to be present (indicated by green bars) has been selected; a few example structures that contain the corresponding property are displayed. For each property, SIRIUS 4 displays the F_1 score that this classifier reached in cross-validation, as well as the number of positive examples in the training data. As expected, most molecular properties are predicted to be absent (indicated by red bars).

SIRIUS 4 can identify the molecular formula of a query compound with very high accuracy; no spectral or structural databases are required for this step of the analysis, as all theoretically possible molecular formulas are considered, allowing one to overcome the limitations of current structure databases. Molecular formulas that are highly atypical for a biomolecule may be penalized (Supplementary Fig. 4) but are never discarded. The CSI:FingerID web service then searches molecular structure databases to identify metabolite structures. CSI:FingerID also provides structural information (namely, the predicted molecular fingerprint) without requiring any molecular structure database, or if no similar structure is present in any database. The predicted fingerprint is prominently featured in the user interface (Fig. 1b–f). In structure-database searches, these may (Supplementary Table 2) or may not[16] be restricted to molecules with known biological relevance, and may also comprise hypothetical metabolites[17]. Hypothetical metabolites and the predicted fingerprint allow SIRIUS 4 to overcome the 'boundaries of known chemical space'. In addition, user-defined structures ('suspects') can be added to the candidate lists. Results are seamlessly integrated into the SIRIUS 4 user interface, which allows users to examine CSI:FingerID's 'reasoning' by a graphical display of predicted molecular properties (Supplementary Fig. 5). Links are provided to structure and spectral databases where a particular compound structure is included.

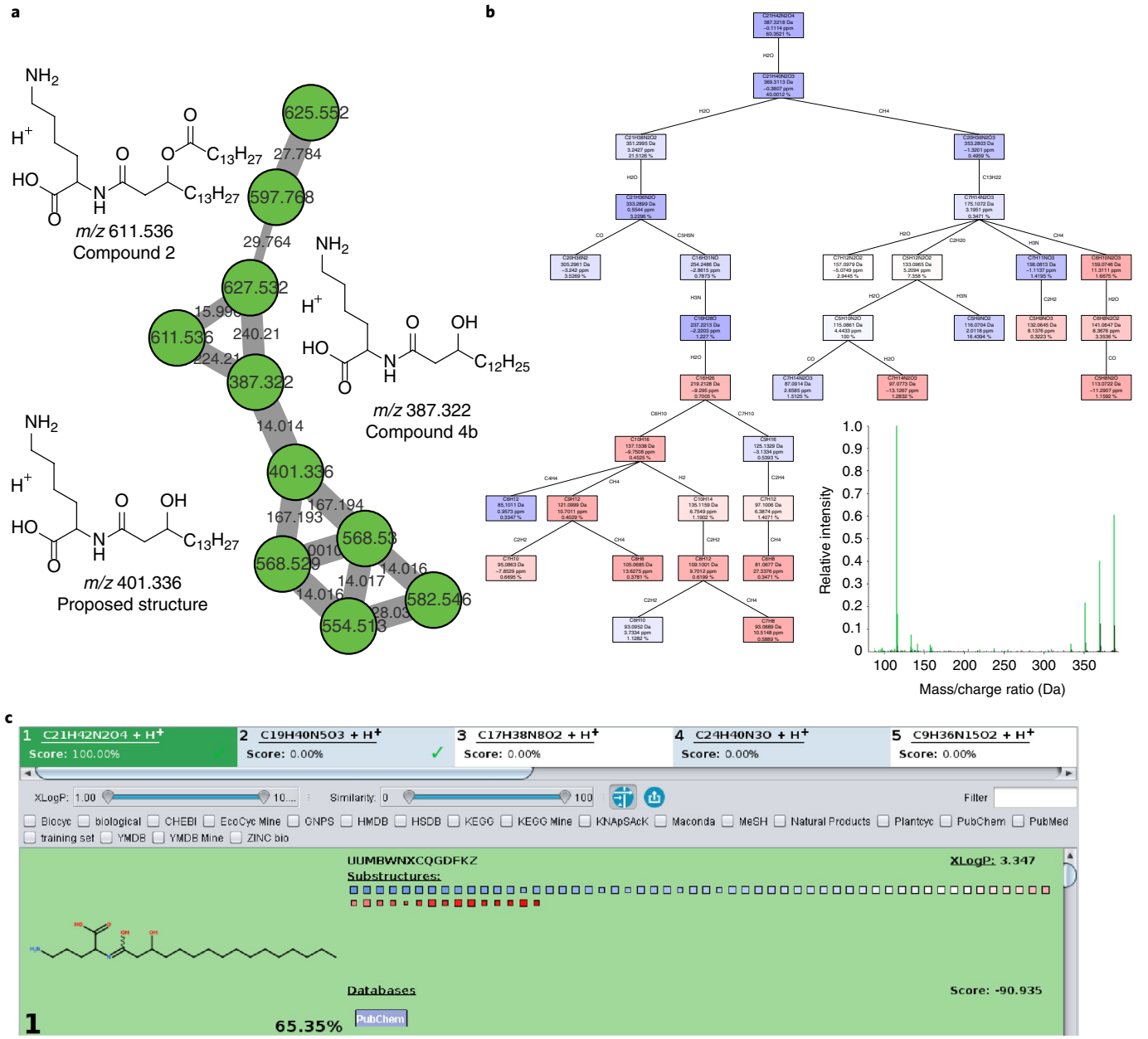

Fig. 2 | In silico annotation of N-acyl amide molecules. a, A network cluster containing *m/z* values matching those of putative N-acyl amide compounds postulated in ref. [21]; none of the compounds could be annotated via library search. **b**, Fragmentation tree that explains the experimentally observed MS/MS fragmentation pattern of the ion with *m/z* 387.322. **c**, Structure of the compound with the highest score, N-3-OH-palmitoyl ornithine, a compound reported in ref. [21] (note enol tautomerism). This structure served as a starting point for annotation of other nodes in the cluster, resulting in a discovery of several N-acyl amides. Structures are labeled according to the original study[21].

To evaluate the isotope pattern scoring algorithm, we compared the performance of SIRIUS 3.0 and SIRIUS 4 using data from 3,965 compounds (Supplementary Fig. 6a, Supplementary Table 3, and Supplementary Results 1). When we used only isotope patterns, correct molecular assignments (top 1) improved by 74.3% (26.6 percentage points) with SIRIUS 4. When we used data from both the first stage of mass spectrometry (MS1) and tandem mass spectrometry (MS/MS), the improvement was still 6.3 percentage points, and the number of wrongly assigned molecular formulas (792 for SIRIUS 3.0, 541 for SIRIUS 4) decreased by 31.7%. We observed improved rates throughout all ranks. For running time evaluation, we again compared SIRIUS 4 with SIRIUS 3.0. We used two datasets, one containing 1,553 compounds (MS/MS only) and the above dataset containing 3,965 compounds (MS1 and MS/MS). For the first dataset, we restricted the set of elements to CHNOPS, to avoid proliferating running times of SIRIUS 3.0 in particular. Here, SIRIUS 4 achieved a 332-fold increase in speed. For the second dataset, we also considered the halogen BrClFI. We ordered compounds by mass and divided them into batches of 50 compounds each. SIRIUS 4 processed all compounds in 5 h 41 min. SIRIUS 3.0 processed the 3,850 lightest compounds in 4 d 1 h, compared with 25 min 4 s for SIRIUS 4, corresponding to a 231-fold increase in speed (Supplementary Fig. 6b and Supplementary Results 2).

Preliminary versions of SIRIUS (3.1.4) and CSI:FingerID were used in the Critical Assessment of Small Molecule Identification (CASMI) 2016 contest[18]. For the 127 compounds with positive ion mode data, CSI:FingerID was the best-performing automated method, reaching 55.1% correct and unambiguous identifications

of structures (ranking the correct structure in first place) when searching ChemSpider[19]. CSI:FingerID showed particularly good performance when independent MS/MS data for the same structure were present in the training data; when MS/MS data of these structures were removed from the training data (structure-disjoint training data; Supplementary Note 2), the rate of unambiguous and correct identifications decreased to 27.6%, which was still better than the rates of all other methods used in CASMI 2016 for positive ion mode data.

We reanalyzed the CASMI 2016 data with the version presented here (Supplementary Table 4 and Supplementary Results 3) and found that for 93.8% of the compounds, SIRIUS 4 assigned the correct molecular formula using MS/MS and isotope pattern data (96.2% in the top 2). When evaluating CSI:FingerID, we ensured structure-disjoint training data. Here, the rate of unambiguous and correct identifications in positive ion mode increased to 39.4% (74.8% in the top 10). For negative ion mode data, we unambiguously and correctly identified 28.4% of the compounds (60.5% in the top 10), again outperforming all other tools. Numbers are similar to or better than the 31.8% correct identifications reported in October 2015 for cross-validation evaluation and positive ion mode[9]. In comparison, the current version reached 40.4% correct identifications in structure-disjoint cross-validation on GNPS, replicating the evaluation setup from ref. [9] (Supplementary Figs. 7 and 8, Supplementary Table 5, and Supplementary Results 4). In practice, it is often reasonable to search in a database focusing on biomolecule structures; searching CASMI 2016 data in a biomolecular structure database with 0.5 million structures (Supplementary Table 2) resulted in 74.0% correct identifications (84.3% in the top 3) for positive ion mode data, again using structure-disjoint cross-validation. For CASMI 2017 category 4 (automatic candidate ranking), SIRIUS 3.5 and CSI:FingerID reached more than six times the number of correct identifications achieved by the best non-CSI:FingerID method[20].

Lastly, we applied SIRIUS 4 to two biological case studies from human fecal and human skin data (Fig. 2, Supplementary Figs. 9–12, and Supplementary Results 5 and 6). In both studies, we investigated structures of molecules that arose as potentially important for understanding biological processes in these studies, but for which no structural elucidations could initially be made. The application of SIRIUS 4 allowed us to determine structures of these key compounds, which we later confirmed by using standards, and thus provided a starting point for further exploration. We use spectral library search and propagation of annotation through molecular networks[1]; analogously, in silico annotations can be propagated through the network. Molecular networking and in silico annotation can be synergistic, but also allow users to verify in silico and network-propagated structure annotations, thus reducing the chance of misannotation.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at https://doi.org/10.1038/s41592-019-0344-8.

Received: 20 March 2018; Accepted: 6 February 2019;
Published online: 18 March 2019

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Acknowledgements

We gratefully acknowledge financial support by the Deutsche Forschungsgemeinschaft (BO 1910/20) to S.B. and the Academy of Finland (310107/MACOME) to J.R.. We thank the GNPS community, S. Stein, F. Kuhlmann, and Agilent Technologies Inc. (Santa Clara, CA, USA) for providing data that were used to estimate the hyperparameters of SIRIUS 4 and to train CSI:FingerID. We also thank F. Kuhlmann and Agilent Technologies for data used to evaluate the isotope scoring.

Author contributions

K.D., P.C.D., J.R., and S.B. designed the research. K.D., M.F., M.L., and J.R. developed computational methods. K.D., M.F., M.L., and M.M. implemented computational methods and performed method evaluations, coordinated by S.B. A.A.A. and A.V.M. performed the biological case studies, coordinated by P.C.D. S.B. wrote the manuscript, to which K.D., M.F., M.L., A.A.A., and A.V.M. contributed, in cooperation with all other authors.

Competing interests

S.B. holds patents (Japanese patent 5559816 and US patent 8263931) whose value might be affected by this publication.

Additional information

Supplementary information is available for this paper at https://doi.org/10.1038/s41592-019-0344-8.

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Methods

Running time improvements, algorithm engineering, and parallelization. Over the past decade, numerous algorithmic improvements and algorithmic engineering advances have resulted in the current fast speed of SIRIUS 4, be it with regard to the decomposition of masses[22,23], the simulation of isotope patterns[6,24], or the computation of fragmentation trees[25,26]. Recall that finding the fragmentation tree that best explains the experimental data is a non-deterministic polynomial time (NP)-hard problem[24], which forbids the existence of a polynomial-time algorithm for this problem, unless P = NP.

Beyond this, SIRIUS 4 uses additional algorithmic tricks to further decrease running times:

- If isotope pattern data are available and at least one molecular formula candidate receives a reasonable positive score, then SIRIUS 4 computes fragmentation trees only for those molecular formula candidates with isotope pattern scores reasonably close to the best score.
- For interpretation of the fragmentation spectrum, at most 60 peaks are considered, sorted by intensity; peaks that have no decomposition are excluded.
- To substantially speed up fragmentation tree computation, SIRIUS 4 proceeds in two rounds. In the first round, it uses a heuristic to get an estimate of the score of a molecular formula candidate. In the second round, only high-scoring molecular formula candidates are processed by an exact algorithm. This ensures exact computation of not only the best solution, but also a number of suboptimal solutions; this is necessary, for example, to make sure that for molecular formula candidates selected by soft thresholding, the optimal fragmentation tree is also computed. Details can be found in ref. [27].
- SIRIUS 4 uses hypothesis-driven recalibration, as described in ref. [8]. For each molecular formula candidate, a fragmentation tree is computed; then, ideal masses from this tree are used to recalibrate the measured mass spectrum. To speed up the recalibration, SIRIUS 4 uses the heuristic fragmentation tree to recalibrate the measured spectrum.
- If the user provides a 'suspect list' of candidate structures, SIRIUS 4 uploads the International Chemical Identifier (InChI) keys of all suspects to the web service and retrieves molecular fingerprints for those present in the structure database. Only for suspects without precomputed fingerprints, SIRIUS 4 computes fingerprints locally. As computing thousands of molecular properties for thousands of candidates is very time consuming, this procedure reduces running times considerably.

Beyond algorithm engineering, SIRIUS 4 now includes a powerful job-scheduling system that further speeds up computations. Practically all present-day processors have multiple cores; SIRIUS 4 automatically detects the number of available cores and uses all of them. Whenever SIRIUS 4 wants to execute some step of the analysis pipeline (computation of fragmentation trees (both heuristically and exactly), web service submission, input/output jobs, and candidate scoring), it pushes the corresponding job to the job-scheduling system. The job-scheduling system decides whether the job can be started immediately (for example, a web service submission where the fragmentation tree has been computed), needs to wait for the completion of other jobs, or is time consuming and needs to wait for a free CPU core. This parallelization is effective across all compounds and analyzed molecular formula candidates of the current workspace. When one of the commercial integer linear programming (ILP) solvers is being used, this parallelization is in effect, and much more efficient than the parallelization offered as part of these ILP solvers. For example, parallelization through the commercial ILP solver on a 16-core CPU will result in a speed increase of, say, threefold, whereas the job scheduler reaches a speedup close to the theoretical optimum as long as enough jobs are to be computed. But when the free GNU Linear Programming Kit (GLPK) solver is used, the implementation of the solver prohibits the simultaneous starting of several instances. Users who rely on the GLPK solver will nevertheless benefit from the new parallelization, as most of the fragmentation trees are computed in parallel by the heuristics, and only a few top-ranking trees have to be computed exactly. The job-scheduling system comes with a job view in the graphical user interface (Supplementary Fig. 2). Individual jobs can be canceled through the job view; dependent jobs will automatically be canceled from the system, too. Similarly, if a job fails, then dependent jobs will not be started. The user can view log files for each of the (failed) jobs individually through the job view. In full, the job scheduling results in a much better workload distribution, in particular on compute clusters with many cores, and is often able to reach CPU loads greater than 90%.

SIRIUS 4 now supports the CPLEX and Gurobi ILP solvers. To allow SIRIUS to be run without installation of a third-party ILP solver, we have integrated the free and open-source GLPK[28] solver. Alternatively, one can use one of the commercial ILP solvers Gurobi (Gurobi Optimization; http://www.gurobi.com) and CPLEX (IBM[29]). Both are free to use for university members, and result in a speedup of up to tenfold for the time-intense exact fragmentation tree computation.

Molecular formula determination. SIRIUS 4 considers all molecular formulas that can possibly explain a certain monoisotopic mass. Senior rules[30] are used to filter out chemically infeasible molecular formulas; beyond this, no filtering is performed. SIRIUS 4 penalizes molecular formulas that are unlikely to correspond to a biomolecule using a linear support vector machine. SIRIUS 4 never discards chemically feasible molecular formulas, and never rewards molecular formulas. SIRIUS 4 also includes a list of 'outlier molecular formulas' that are atypical for a biomolecule but are not penalized, such as $C_{10}HF_{19}O_2$ for perfluorodecanoic acid. SIRIUS 4 can consider numerous ion types; for ion type 'unknown', SIRIUS will consider ion types protonation, sodium, and potassium for positive ion mode, and deprotonation and chlorine for negative ion mode. For a combined search with CSI:FingerID, SIRIUS 4 can be restricted to molecular formulas that are found in a structure database such as PubChem. Further details are provided in Supplementary Note 3.

SIRIUS 4 uses the maximum a posteriori estimation from ref. [8] for the computation of fragmentation trees. Compared with molecular formula identifications obtained with SIRIUS versions before version 3.0, this results in massive improvements: evaluations in Fig. 4 of ref. [8] show that molecular formula identification rates practically double compared with those of earlier versions when relying solely on MS/MS data. Böcker and Dührkop[8] argue that the improved performance is due to an increase in fragmentation tree structural quality. High fragmentation tree quality is, in turn, required for a successful structural identification via CSI:FingerID.

Isotope pattern analysis. SIRIUS 4 implements a scoring for comparing experimental and simulated (ideal, theoretical) isotope patterns that differs from, and improves on, previous versions. Mass deviations and mass difference deviations are still modeled by normal distributions[6]. For intensities, previous models consider only either relative errors[6] or absolute errors[31], and cannot deal with both types simultaneously. The model from ref. [32] tries to overcome this limitation, but it is not intuitive and requires many parameters.

To simultaneously model absolute and relative intensity deviations, we propose a simple maximum likelihood estimator that requires only two parameters, namely, absolute error ($\sigma_{abs} > 0$) and relative error ($\sigma_{rel} > 0$) of peak intensities. We statistically model the intensity error as

$$Y = x + D + E$$

where x is the expected (theoretical) intensity, Y is the random variable modeling the observed intensity, and D and E are random variables for relative and absolute noise, respectively. We assume that both relative noise, $D \sim x\,\mathcal{N}(0, \sigma_{rel}^2) = \mathcal{N}(0, x^2\sigma_{rel}^2)$, and absolute noise, $E \sim \mathcal{N}(0, \sigma_{abs}^2)$, are normally distributed. In detail, we assume that D and E have densities

$$f_D(\delta) = \frac{1}{\sqrt{2\pi x^2 \sigma_{rel}^2}} \exp\left(-\frac{\delta^2}{2x^2\sigma_{rel}^2}\right)$$

$$f_E(\epsilon) = \frac{1}{\sqrt{2\pi\sigma_{abs}^2}} \exp\left(-\frac{\epsilon^2}{2\sigma_{abs}^2}\right)$$

for $\delta, \epsilon \in \mathbb{R}$. We are using the probability density function to estimate these probabilities, which can be interpreted as the limit of an arbitrary small interval around the values δ and ϵ, respectively. Note that this model can result in negative observed peak intensities; we found that this limitation is not relevant in application, where relatively weak noise is observed.

We further assume that relative and absolute noise are independent. Given an observed intensity y and an expected intensity x, the likelihood of some model $\theta = (\delta, \epsilon)$ is

$$\begin{aligned} \mathcal{L}_{y|x}(\delta, \epsilon) &= \frac{1}{\sqrt{2\pi}\, x\sigma_{rel}} \exp\left(\frac{-\delta^2}{2x^2\sigma_{rel}^2}\right) \times \frac{1}{\sqrt{2\pi}\,\sigma_{abs}} \exp\left(\frac{-\epsilon^2}{2\sigma_{abs}^2}\right) \\ &= \frac{1}{2\pi x \sigma_{rel}\sigma_{abs}} \exp\left(-\frac{\delta^2}{2x^2\sigma_{rel}^2} - \frac{\epsilon^2}{2\sigma_{abs}^2}\right) \end{aligned} \tag{1}$$

We find that the maximum likelihood $\mathcal{L}_{y|x}(\delta_0, \epsilon_0)$ for model (δ_0, ϵ_0) is

$$\mathcal{L}_{y|x}(\delta_0, \epsilon_0) = \frac{1}{2\pi x \sigma_{rel}\sigma_{abs}} \exp\left(-\frac{(y-x)^2}{2(\sigma_{abs}^2 + x^2\sigma_{rel}^2)}\right) \tag{2}$$

(Supplementary Note 4). But this implies that for the computation of the maximum likelihood, we actually do not have to compute δ_0 or ϵ_0; instead, it is sufficient to directly insert the theoretical intensity x and observed intensity y into equation (2) to estimate the maximum likelihood of the data. We note that equation (2) is very similar to the probability density function of the random variable $D + E \sim \mathcal{N}(0, \sigma_{abs}^2 + x^2\sigma_{rel}^2)$.

The above model is mathematically sound, but it has a conceptual disadvantage when a set of candidate isotope patterns is being scored against one measured isotope pattern: the relative noise depends on the peak intensity in the theoretical (candidate) isotope pattern, and hence each candidate pattern is scored differently. To this end, we have exchanged the roles of x and y (expected/theoretical intensity versus observed intensity) in our implementation of the scoring for SIRIUS 4. Likelihoods computed in this way are usually very large, positive values. SIRIUS 4

uses log odds, dividing values through the likelihood for $\delta_0 = 2\sigma_{rel}$ and $\epsilon_0 = 2\sigma_{abs}$. By default, SIRIUS 4 uses parameters $\sigma_{abs} = 0.01$ and $\sigma_{rel} = 0.08$.

For a candidate molecular formula, we simulate an isotope pattern with peak intensities and mean peak masses as described in refs. [6,24]. We normalize both spectra using the first isotope peak. Assuming statistical independence[6,24], the likelihood of the candidate molecular formula is simply the product of the individual likelihoods for peak mass differences (modeled by normal distributions) and peak intensity differences (modeled above). For peaks that are not observed but expected in the theoretical pattern, we estimate no likelihood for mass deviation, but a likelihood for intensity deviation where the observed intensity is set to zero. We extract isotope patterns from MS1 by searching for the most intensive peak around the precursor mass within the allowed mass deviation; then, we extend the pattern gradually by picking the next isotope peak that has a reasonable mass difference. If several such peaks exist, they are merged, using the weighted mean of their masses and the sum of their intensities. To ensure that peaks that are not part of the isotope pattern (for example, coeluting ions) are not accidentally added to the pattern, we compute scores for all possible lengths of the patterns by successively removing the last peak of the pattern, and report only the maximum score.

Isotope pattern analysis for MS/MS data. In some experimental setups ('all ion fragmentation' and so on), isotope peaks and fragment peaks are measured together in the same spectrum. For such experiments, SIRIUS 4 offers a combined isotope and fragmentation pattern analysis. Details are provided in Supplementary Note 5.

Automated detection of uncommon elements. SIRIUS 4 integrates automated detection of 'uncommon elements' from the isotope pattern of the query compound. These elements are added to the standard set of elements CHNOP when the molecular formula of the query compound is determined. Even if a particular element is not excluded at this step, the subsequent analysis can still choose a molecular formula that contains zero atoms of the element. Isotope pattern analysis can be used to exclude the elements sulfur, chlorine, bromine, boron, and selenium, as these have characteristic isotope patterns. There is also a predictor for silicon that is disabled by default, as it results in a relatively large number of false positives. Different from ref. [14], we use a DNN[33] for this task. In this way, we reduce the memory requirements of the uncommon-element prediction from more than 200 MB for random forests[14] to 75 KB for the DNN. Details are provided in Supplementary Note 6.

SIRIUS 4 step-by-step computational workflow. We now describe in detail the steps that SIRIUS 4 performs when analyzing the data from a single compound. For each compound, SIRIUS 4 inputs an MS/MS spectrum and, optionally, an isotope pattern in MS1. SIRIUS 4 aims to identify the molecular formula of the query compound and annotate the MS/MS spectrum with a fragmentation tree. Dührkop et al.[27] noted that fragmentation trees computed by heuristics can have structures that deviate notably from that of the optimal solution; for this reason, we want to ensure that all fragmentation trees of the top-scoring candidate molecular formulas are computed exactly. This is based on the general assumption that the structure of the optimum solution is closer to the 'true' structure than some arbitrary suboptimal solution.

In detail, SIRIUS 4 proceeds as follows:

1. If an isotope pattern is provided, we use the DNN from Supplementary Note 6 to restrict the set of elements used.
2. We enumerate over all molecular formulas that explain the precursor ion peak, or (if provided) the monoisotopic peak of the isotope pattern, using the Round-Robin algorithm[22,23].
3. If an isotope pattern is given, we score each of these molecular formulas using the maximum likelihood scoring described above. If at least one isotope pattern receives a reasonable score, we discard all molecular formula candidates with very low scores.
4. For each molecular formula candidate, we do the following: create a fragmentation graph from the MS/MS data rooted in the molecular formula candidate. Edge weights in this graph are chosen as described in ref. [8]. Different from that procedure, we use the support vector machine (SVM) from Supplementary Note 3 to penalize outlier molecular formulas. If an MS1 isotope pattern is given, we modify the weight of the root by the maximum likelihood score from the isotope pattern analysis. For the nodes of the graph, we decompose the 60 most intense peaks in the MS/MS spectrum, enumerating all molecular formula explanations.
5. If in-source fragments are detected in the MS1, we use their isotope score to modify the score of the corresponding nodes in the fragmentation graph.
6. For each molecular formula candidate and corresponding fragmentation graph, we compute a fragmentation tree (colorful subtree) using the Critical Path[3] heuristic from ref. [27].
7. We sort the molecular formula candidates according to their scores.
8. For the top k molecular formula candidate and corresponding fragmentation trees:
   (a) we use hypothesis-driven recalibration[8,34] to recalibrate the MS/MS spectrum (see ref. [8]),
   (b) we compute the fragmentation graph and edge weights as described above, but using the recalibrated MS/MS spectrum, and
   (c) we compute the optimum fragmentation tree (maximum colorful subtree) using an exact algorithm, namely, the integer linear program from ref. [25].
9. We sort these k fragmentation trees according to their new scores; top-scoring trees are reported to the user.

By default, SIRIUS 4 computes $k = 20$ fragmentation trees exactly.

Integrating CSI:FingerID as a web service. The integration of CSI:FingerID into SIRIUS is realized by means of a RESTful web service via HTTPS, and implemented as a Java servlet. The deliberately simple client-server architecture enables high performance, reliability, and scalability. Running CSI:FingerID as a web service avoids unnecessary maintenance and incompatibilities through upgrading, and does not require third-party libraries on the user side. Furthermore, CSI:FingerID is in a stage of rapid methodical progress, and the chosen architecture allows us to continuously integrate methodical upgrades without the user having to install new releases. No spectral libraries or compound structure databases have to be installed or updated on the user side, and additional training data and structure databases can also be integrated continuously, without the need for upgrades. This is particularly relevant for the integration of spectral libraries that are freely available for training computational methods, but not for download.

The workflow for the integration of CSI:FingerID and SIRIUS is depicted in Fig. 1a. The fragmentation tree and spectrum are uploaded from the SIRIUS client to the web server and temporarily stored in a relational job database. A compute cluster is regularly fetching jobs from the database, predicting fingerprints, and storing results in the database. The SIRIUS client downloads these predicted fingerprints, as well as a list of candidate structures and fingerprints, via the REST application programming interface. Processing of a single compound instance is performed by the web service in less than 1 s of wall-clock time, enabling an interactive analysis of the data without disturbing delays. Scoring, ranking, and visualization of the candidate structures are done locally on the client side.

The strict separation between web interface and compute cluster allows us to easily scale the web service: in times of high workload (that is, when more than 50 jobs are stored in the job database), additional compute nodes are spawned on the compute cluster. Each node that has access to the job database can become a compute node.

To ensure that the CSI:FingerID web service is available at all times, we have installed the complete setup on two independent physical machines. The two physical machines are permanently monitored; if one machine is no longer responding to requests of the CSI:FingerID web API, the second machine takes over. Furthermore, we now give users a time window for upgrades of SIRIUS installations; previously, any change to the web application programming interface required that all users immediately upgrade their SIRIUS installations, as the old web API would no longer respond. Starting with SIRIUS 4, we can run several versions of the CSI:FingerID API in parallel. For future updates, the outdated version will work for at least a week, to give users time for upgrading.

Negative ion mode support for CSI:FingerID. Until recently, CSI:FingerID did not support the analysis of negative ion mode spectra, because of the lack of publicly available training data. With the integration of negative ion mode spectra from the National Institute of Standards and Technology (NIST) (see below), CSI:FingerID 1.1 can now analyze negative ion mode spectra. Positive and negative ion mode spectra are trained separately. With the exception of the multiple kernel learning weights (Supplementary Note 7), the integration of negative ion mode did not require any changes to the CSI:FingerID method.

Kernels for CSI:FingerID. Compared with the previous version, CSI:FingerID 1.0, we integrated ten additional kernels into the support vector machine used for predicting fingerprints, and removed old ones that did not contribute to the search performance of CSI:FingerID. CSI:FingerID 1.1 uses the kernels described in Supplementary Table 1; kernel weights are computed using the ALIGNF[35] and ALIGNF+ (ref. [36]) multiple kernel learning methods.

Additional training data for CSI:FingerID. We have integrated new training data into the CSI:FingerID web service. Next to spectra from MassBank[37] (https://massbank.eu/) and GNPS[1] (https://gnps.ucsd.edu/), we trained CSI:FingerID on 16,858 compounds from the NIST 2017 database (commercial; NIST, v17). In full, CSI:FingerID 1.1 is trained on 16,083 structures, with 19,118 independent MS/MS measurements (compounds) in positive mode and 10,823 measurements in negative mode. We observed that the prediction performance of CSI:FingerID decreases when it is trained on all available spectra from GNPS and MassBank. Therefore, we train CSI:FingerID in a two-step approach that removes outliers[38], using a tenfold cross-validation. In this way, we discard 757 structures. Further details are provided in Supplementary Note 8.

Molecular properties for CSI:FingerID. For CSI:FingerID 1.1, we have added extended-connectivity fingerprints (ECFPs) to the list of predicted molecular properties[39]. We use molecular properties from ECFP6 that are found sufficiently

often in the training structures. Molecular properties are computed using the Chemistry Development Kit[40]. In total, 7,593 molecular properties are available for learning. We have found that 2,937 molecular properties in positive and 1,996 in negative ion mode can be learned sufficiently well from the training data. Details are provided in Supplementary Note 9.

Candidate retrieval and adduct types for CSI:FingerID. SIRIUS 4 considers all structure candidates from the selected structure database that agree with the molecular formula of the parent ion determined in the previous analysis step. To ensure that candidates with slightly suboptimal molecular formulas are considered in this search, SIRIUS 4 uses a 'soft threshold': all molecular formula candidates of the parent ion with scores above 0.75 of the optimal score are considered.

When retrieving candidates, SIRIUS 4 can consider numerous adduct types, as well as in-source fragments—for example, $[M+H]^+$ for protonation; $[M]^+$ for intrinsically charged molecules; $[M-H_2O+H]^+$ for in-source fragments; and $[M+NH_4]^+$, $[M+CH_4O]^+$, $[M+C_2H_3N]^+$, $[M+C_4H_6N_2]^+$, and so on for adducts. Note that 'ion types' (protonation, sodium, potassium, deprotonation, chlorine) have already been selected for molecular formula determination (Supplementary Note 3). In the graphical user interface, the user can select or unselect all of these adduct types and in-source fragments individually. For example, if SIRIUS 4 determines the molecular formula of the parent ion as $C_{17}H_{25}N_2O_4^+$ and adduct type $[M+NH_4]^+$ (ammonium adduct) is selected, then candidate structures with molecular formula $C_{17}H_{21}NO_4$ are retrieved from the structure database; the union of these candidate structures is scored against the predicted fingerprint, sorted, and displayed by SIRIUS 4. Furthermore, SIRIUS 4 can consider user-defined adduct types.

Tree-based maximum a posteriori estimation as a CSI:FingerID score. For searching in molecular structure databases, we have integrated a posterior probability score[15] that outperforms all scores from the original CSI:FingerID release[9]. The score models dependencies between molecular properties when scoring candidate structures. Details are provided in Supplementary Note 10 and ref. [15].

Structure databases. We have updated our PubChem database copy used for searching; the current version (downloaded 13 August 2017) contains 93,859,798 compounds and 73,444,774 unique structures, constituting a 77.3% increase in compounds and an 80.0% increase in structures over those in ref. [9]. For our biomolecule structure database, we combined compound structures from numerous public databases (Supplementary Table 2). The combined database contains 492,921 unique structures of biomolecules and compounds that can be expected in biological samples. Furthermore, CSI:FingerID allows the user to search the Metabolic In Silico Network Expansion databases[17] (EcoCyc, Yeast Metabolome Database, Kyoto Encyclopedia of Genes and Genomes) of 651,824 in silico–generated structures. Finally, users can provide a suspect list of structures as a text file of InChI strings, Simplified molecular-input line-entry system string or Chemical table files. This suspect list can be searched in addition to, say, the biomolecule structure database. The numbers reported above represent only snapshots, as structure databases are continuously updated and extended.

Graphical user interface. A usual analysis proceeds as follows (Supplementary Fig. 1): mass spectrometry data can be loaded into SIRIUS 4 via a drag-and-drop of one or more files. In the first step, isotope pattern and MS/MS data are used to determine the molecular formula of the compound, and to compute a fragmentation tree. In this step, SIRIUS 4 will usually not limit its computations to molecular formulas present in any database, and considers all molecular formulas that can possibly explain the observed precursor mass. Molecular formula candidates and their scores (isotope pattern, fragmentation tree, and total) are displayed in the 'SIRIUS Overview' tab. For the total score, SIRIUS 4 reports the posterior probability of the molecular formula given the data; this must not be mistaken for the probability that the molecular formula is correct. A blue line indicates the soft score threshold; by default, only molecular formulas with scores above this threshold are used for searching with CSI:FingerID. Two tabs allow an in-depth inspection of the spectra data and the fragmentation tree for each molecular formula candidate.

The user can then initiate a molecular structure search. The corresponding fragmentation trees and spectra are uploaded to the CSI:FingerID web server, and results are retrieved and, finally, displayed in the user interface. By default, SIRIUS 4 selects one or more molecular formulas for the downstream analysis on the basis of a soft threshold, but the user can override this choice. The 'CSI:FingerID Overview' tab summarizes results over all molecular formulas. As the predicted fingerprint differs depending on the candidate molecular formula, the 'CSI:FingerID Details' tab allows the user to browse through molecular formula candidates individually; for the best-scoring candidate, the molecular formula is highlighted in green. The tab also shows which predicted molecular features (presence or absence of certain substructures) support a particular candidate structure, and which predicted molecular features contradict that candidate. True positive predictions (molecular properties that are presented in the candidate, in agreement with the predicted fingerprint (blue)) and false negative predictions (molecular properties that are presented in the candidate but not in the predicted fingerprint (red)) are displayed as boxes. The size of a box indicates the prediction quality (F_1 score) for this molecular property in cross-validation, whereas color and shade indicate how sure CSI:FingerID is about the prediction for this particular compound (Platt probability). A mouse-over displays the SMARTS string encoding the molecular property; clicking on the square highlights the corresponding substructure in the candidate structure. Results can be filtered on the basis of XlogP values[41] that are predicted using the Chemistry Development Kit[40,42]. SIRIUS 4 also displays all databases where each structure candidate can be found; clicking on the database name directly opens the corresponding website in the browser.

Finally, the 'Predicted Fingerprint' tab allows the user to view the fingerprint that was predicted by CSI:FingerID for the query compound, for each molecular formula candidate (Fig. 1 and Supplementary Fig. 5). For any property, the user interface displays graphical examples of the predicted substructures. Conceptually, fingerprint prediction is executed before a search in a structure database, and it is not required that the query compound be contained in any structure database for this fingerprint to be predicted. In fact, predicted molecular properties can be used to hypothesize about the structure of a compound that is not present in any database.

The user interface offers additional filtering options, but also full text search for every output list to, say, search for a particular InChI string. Details can be found in the SIRIUS manual[43].

Biological case studies. A step-by-step protocol for the biological case studies (Supplementary Results 5 and 6) is available as a Supplementary Protocol and at Protocol Exchange[44].

For biological case study 1 (fecal), the publicly available data were used for the analysis. For biological study 2 (skin), all relevant ethical regulations were complied with in accordance with the study (Institutional Review Board 161694) approved by the University of California, San Diego, Human Research Protections Program.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Code availability

SIRIUS 4 is written in Java; is open source under the GNU General Public License (version 3); and works on Windows, macOS X, and Linux. In addition to the graphical front end, a comprehensive command-line version allows batch processing and integration into workflows; integration into GNPS[1], OpenMS[2], and MZmine[4] is ongoing. We also provide source code, executable binaries, documentation, support, non-commercial training data, example files, and additional information on the SIRIUS website (https://bio.informatik.uni-jena.de/sirius/); a source copy is hosted on GitHub (https://github.com/boecker-lab/sirius). You can retrieve the InChIs of all compounds used to train CSI:FingerID from the web service (https://www.csi-fingerid.uni-jena.de/webapi/trainingstructures.csv?predictor=pos and https://www.csi-fingerid.uni-jena.de/webapi/trainingstructures.csv?predictor=neg).

Data availability

Data for the CASMI 2016 re-evaluation are available from https://bio.informatik.uni-jena.de/data under a Creative Commons CC-BY license. Cross-validation data for the GNPS search re-evaluation are available from https://bio.informatik.uni-jena.de/data/ (Creative Commons CC0 1.0 Universal license). Data for the American Gut project are available in the MassIVE database (MSV000080186 and MSV000080187; Creative Commons CC0 1.0 Universal license). The analysis can be accessed via the GNPS website (http://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=9bd16822c8d448f59a03e6cc8f017f43 and http://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=d26ae082b1154f73ac050796fcaa6bda). Data for the study of clothing with antibacterial properties are available at MassIVE (MSV000081379; Creative Commons CC0 1.0 Universal license). Analysis is available at the GNPS website (https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=a5e8ca1b7a9c42cfb45fbb2855e36721). Source data for Supplementary Figs. 6–8 are available online.

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Our web collection on statistics for biologists may be useful.

Software and code

Policy information about availability of computer code

Data collection	All data used in this study was collected from open & public repositories, as detailed in the manuscript.
Data analysis	See Software statement; SIRIUS 4.0.1 is available from GitHub and uses the GPL license.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

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Data for the CASMI 2016 reevaluation are available from https://bio.informatik.uni-jena.de/data under the Creative Commons CC-BY license. Cross validation data for the GNPS search reevaluation are available from https://bio.informatik.uni-jena.de/data/ (Creative Commons CC0 1.0 Universal license). Data for the American Gut project are available from https://massive.ucsd.edu/, data sets MSV000080186 and MSV000080187 (Creative Commons CC0 1.0 Universal license). The analysis

can be accessed via http://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=9bd16822c8d448f59a03e6cc8f017f43 and http://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=d26ae082b1154f73ac050796fcaa6bda. Data for the clothing with antibacterial properties study are available at https://massive.ucsd.edu/, data set MSV000081379 (Creative Commons CC0 1.0 Universal license). Analysis at https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=a5e8ca1b7a9c42cfb45fbb2855e36721.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	For the biological case study #1 (fecal), no sample size calculation was carried out. The study was exploratory in nature, so no observation of specific effect was a priori anticipated. The discovery of novel metabolites did not rely on statistical significance and thus the sample size is not relevant as long as the detection could be reliably established across multiple samples. For the biological study #2 (skin), an equal number of participants of both genders (6 men and 6 women) were chosen to avoid gender bias. The study design was 12 N=1 studies where changes within an individual over time upon interventions were monitored as opposed to a cohort. The duration of the study was such to longitudinally collect samples over the course of skin cells turnover (typically 4 to 6 weeks).
Data exclusions	No data were excluded in both biological case studies.
Replication	The detection of compounds was confirmed across multiple independent samples in each study and the structural identification was confirmed using authentic standard when available.
Randomization	In both biological case studies samples were randomized during MS analysis to minimize possible batch effect.
Blinding	In both biological case studies the detection of reported molecules does not rely on data allocation and thus blinding is not consequential.

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☒	☐ Unique biological materials
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☐	☒ Human research participants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

Policy information about studies involving human research participants

Population characteristics	For both biological case studies participants were healthy adults aged 20 to 69 years with body mass indexes (BMIs) ranging between 18.5 and 30 kg/m2. The genders were equally matched for the biological study #2 (skin).
Recruitment	For the biological case study #1 (fecal), the participants were recruited via the American Gut project infrastructure (https://fundrazr.com/americangut?ref=ab_5hzRHYfmoYR5hzRHYfmoYR). For the biological study #2 (skin), a solicitation email according to the study IRB has been sent to potential participants. The volunteers were enlisted into study on the first-come, first-serve basis.